1.2.2 Use of apparatus and techniques

	Learning outcomes	Additional guidance
	Through use of the apparatus and techniques listed below, and a minimum of 12 assessed practicals (see Section 5h), learners should be able to demonstrate all of the practical skills listed within 1.2.1 and CPAC (Section 5h, Table 2) as exemplified through:	
(a)	use of appropriate analogue apparatus to record a range of measurements (to include length/distance, temperature, pressure, force, angles and volume) and to interpolate between scale markings	HSW4
(b)	use of appropriate digital instruments, including electrical multimeters, to obtain a range of measurements (to include time, current, voltage, resistance and mass)	HSW4
(c)	use of methods to increase accuracy of measurements, such as timing over multiple oscillations, or use of fiducial marker, set square or plumb line	HSW4
(d)	use of a stopwatch or light gates for timing	HSW4
(e)	use of calipers and micrometers for small distances, using digital or vernier scales	HSW4
(f)	correctly constructing circuits from circuit diagrams using DC power supplies, cells, and a range of circuit components, including those where polarity is important	HSW4
(g)	designing, constructing and checking circuits using DC power supplies, cells, and a range of circuit components	HSW4
(h)	use of a signal generator and oscilloscope, including volts/division and time-base	HSW4
(i)	generating and measuring waves, using microphone and loudspeaker, or ripple tank, or vibration transducer, or microwave/radio wave source	HSW4

(j)	use of a laser or light source to investigate characteristics of light, including interference and diffraction	HSW4
(k)	use of ICT such as computer modelling, or data logger with a variety of sensors to collect data, or use of software to process data	HSW3, HSW4
(l)	use of ionising radiation, including detectors.	HSW4